MUKUL VED

Data Engineer | Machine Learning Engineer | Analytics Engineer

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PROFESSIONAL SUMMARY

Data and Machine Learning Engineer with 3+ years of experience building production-grade batch and streaming pipelines, cloud data lakes, and end-to-end ML systems across enterprise technology, professional services, and non-profit domains. Engineered PySpark, Kafka, and Airflow pipelines processing 25M+ records monthly, cut complex query latency from 22 seconds to under 7 seconds, and shipped predictive and NLP models with SMOTE class balancing and SHAP interpretability into live decision workflows. Fluent in Python, SQL, PySpark, dbt, and AWS/Azure, with a track record of compliance-ready design (GDPR, HIPAA, FERPA) and 40+ BI dashboards influencing $68M+ in revenue decisions.

TECHNICAL SKILLS

Programming and Scripting: Python, SQL, R, JavaScript (ES6+), Bash, PL/pgSQL

Data Engineering and Big Data: PySpark, Apache Spark, Apache Kafka, Apache Airflow (DAG authoring, retries, SLAs), dbt, Hadoop (HDFS, HBase), ETL and ELT, Batch and Streaming Pipelines, Data Lake and Data Warehouse Architecture, Data Modeling (Star and Snowflake schema), Partitioning, Indexing, Incremental Ingestion, Materialized Views, Schema Alignment, Data Quality and Validation

Machine Learning and AI: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Logistic Regression, Random Forest, Gradient Boosting, SMOTE and Class Imbalance Handling, Feature Engineering, Cross-Validation, Model Evaluation (Precision, Recall, F1, AUC-ROC), SHAP and Feature Importance, Model Deployment and Retraining Workflows

NLP and LLMs: spaCy, NLTK, VADER Sentiment Analysis, LDA Topic Modeling, TF-IDF, Tokenization and Lemmatization, Large Language Models (LLMs), Prompt Engineering, Azure OpenAI, LangGraph, Multi-Agent Systems

Cloud and Infrastructure: AWS (S3, SageMaker, Lambda, RDS), Microsoft Azure (Azure OpenAI Service), Google Cloud (BigQuery), Snowflake, Docker, Git, CI/CD, REST APIs, FastAPI, Flask

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Neo4j (graph), PostGIS

Analytics and BI: Tableau, Power BI, Looker, Excel (advanced modeling), Matplotlib, Seaborn, Streamlit, A/B Testing, Hypothesis Testing, ANOVA, Time Series Analysis

Governance and Compliance: GDPR, HIPAA, FERPA, Data Retention Policies, Role-Based Access Control, Encryption at Rest and in Transit, Audit Logging, Data Anonymization

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Chicago Education Advocacy Cooperative (ChiEAC), Chicago, IL **Apr 2025 - Present**

- Built an early-warning ML system that flags student disengagement risk across 12 partner schools, training Logistic Regression, Random Forest, and Gradient Boosting classifiers in Python and scikit-learn on 10,000+ internal survey responses joined with public socioeconomic datasets.
- Raised recall on the minority at-risk class by applying SMOTE oversampling and weighted loss functions, benchmarking models with cross-validation on precision, recall, and F1 to minimize false negatives on students who were truly disengaged.
- Drove educator adoption of model outputs by generating SHAP values and feature-importance explanations, converting opaque risk scores into the specific drivers behind each flag.
- Engineered modular, reusable preprocessing pipelines in Pandas and NumPy (Min-Max and z-score scaling, one-hot encoding, mean and contextual imputation, tokenization) that made 5 longitudinal cohorts machine-learning ready and reproducible across research partners.
- Surfaced hidden behavioral signals in 3,500+ open-text survey responses by building an NLP pipeline with spaCy, NLTK, and scikit-learn using VADER sentiment scoring, LDA topic modeling, and TF-IDF vectorization, then correlated sentiment polarity with model-predicted dropout risk.
- Informed curriculum redesign across 12 schools by delivering sentiment and risk dashboards (Matplotlib, Seaborn) with cohort and school-level drill-downs, built under FERPA-aligned anonymization practices.

Data Analyst Intern | Excelrate.org, Remote, USA **Aug 2023 - Oct 2023**

- Cut annual advertising spend by $42K and expanded campaign reach into 5 new regions by directing a 16-member cross-functional team through a granular SQL and Excel spend analysis benchmarking CPC, CPA, and ROAS across campaigns and regions.
- Reduced reporting cycles by 14 hours per month by designing automated Tableau dashboards wired directly to SQL databases with real-time refresh, filters, and drill-down, replacing manually assembled static Excel reports.
- Reallocated $18K toward high-performing campaigns by partnering with HR and finance to integrate payroll-linked expenses into campaign cost models, exposing true fully loaded cost per campaign for the first time.

Data Analyst | Dell Technologies, New Delhi, India **Jun 2022 - May 2023**

- Enabled near real-time operational analytics on 25M+ device telemetry and support ticket records per month by building hybrid batch and streaming pipelines with PySpark, Apache Kafka, and Apache Airflow, replacing batch-only workflows that delayed insight by hours.
- Eliminated pipeline data loss and reprocessing cost by implementing checkpointing, retry mechanisms, region and device-type partitioning, and incremental ingestion across high-velocity ingestion spikes from 190+ global regions.
- Established a single source of truth for global IT assets by consolidating AWS S3 and on-prem SQL Server datasets into a unified data lake, applying schema alignment, metadata normalization, and timestamp standardization across 190+ regions.

- Reduced complex query latency by 68%, from 22 seconds to under 7 seconds, through SQL tuning (indexing, partitioning, query restructuring), materialized views, Pandas in-memory transformations, and modular dbt models.
- Influenced $68M+ in support revenue by modernizing revenue and warranty analytics workflows in SQL, Pandas, and dbt, building reusable dbt models for renewal rates, replacement cycles, and contract lifecycle stages, and correlating telemetry signals with warranty events to enable proactive customer outreach.
- Reduced regulatory exposure on 12M+ sensitive customer and device records by designing GDPR and HIPAA-aligned storage in PostgreSQL and MongoDB with encryption at rest and in transit, role-based access control, audit logging, and Airflow-scheduled retention and archival jobs.
- Improved global asset utilization by delivering 15 Tableau and Power BI dashboards on supply chain flows, warranty claims, and capacity forecasts, giving operations managers drill-down views to reallocate inventory across top product markets.

Data Analyst | KPMG, Gurugram, India **May 2021 - May 2022**

- Reduced audit reporting latency from hours to under 5 seconds by engineering distributed ETL pipelines with PySpark and Apache Airflow DAGs that ingested and transformed 18M+ financial and operational records monthly into a centralized warehouse.
- Cut pipeline failure and manual troubleshooting by configuring Airflow dependency management, incremental loads, error handling, retry logic, and job-level logging across concurrent client engagements.
- Lifted client satisfaction scoring by 1.8 points by synthesizing fragmented survey and feedback datasets in Python, Pandas, and SQL, and applying A/B testing and ANOVA to statistically validate which engagement interventions actually moved the needle.
- Shortened executive decision cycles by 2.4 days by designing 12 interactive Power BI and Tableau dashboards on regional revenue, workforce utilization, and compliance KPIs, connected directly to the centralized warehouse and data lake.
- Reduced escalation cases by 1,200 annually by modeling structured project financials in PostgreSQL alongside unstructured consultant notes and tickets in MongoDB, applying anomaly detection and NLP to flag accounts likely to escalate, with feature-importance outputs so managers understood why.
- Improved audit reporting efficiency by 15 hours per cycle by orchestrating an AWS S3 data lake with dbt transformations, access controls, and audit logging, and scheduling Airflow refreshes so compliance teams stopped rebuilding evidence packs by hand.

SELECTED PROJECTS

PromptRx: LLM Discharge Instruction Simplification | *Azure OpenAI (GPT-4.1), Flask, Streamlit, Python*

- Built an LLM-driven system that converts dense ER discharge instructions into plain-language, patient-friendly summaries auto-categorized into emergency signs, diagnosis, medications, home care and restrictions, follow-ups, and unclear instructions, addressing the 70%+ of patients who misunderstand discharge guidance.
- Designed a HIPAA-aware architecture on Azure OpenAI Service with a factual validation harness that diffs original versus simplified instructions to confirm no clinical content is dropped, plus multilingual output for non-English-speaking patients.

Insurance Claims Fraud Detection | *Neo4j, XGBoost, Python*

- Achieved 0.91 AUC-ROC on 2M+ claims by combining graph features from a 200K-node Neo4j network (provider, claimant, and policy relationships) with XGBoost gradient boosting to surface coordinated fraud rings that flat tabular models miss.

Prior-Authorization Decision Agent | *LangGraph, FastAPI, Python*

- Built a multi-agent system that automates prior-authorization decisioning, orchestrating retrieval, policy evaluation, and decision agents through a LangGraph state machine exposed over a FastAPI service.

Real-Time Opioid Overdose Surveillance | *Apache Kafka, LightGBM, Python*

- Streamed surveillance signals through Kafka into a LightGBM risk model reaching 0.83 AUC-ROC, enabling near real-time detection of emerging overdose clusters rather than retrospective reporting.

Hospital Price Transparency Engine | *Snowflake, dbt, Python*

- Normalized wildly inconsistent price-transparency files from 500 hospitals into a queryable Snowflake warehouse using dbt transformations, achieving 94%+ schema-mapping accuracy across heterogeneous source formats.

SDOH Risk Scoring Engine | *XGBoost, PostGIS, PostgreSQL*

- Engineered a social-determinants-of-health risk score by joining 8 federal datasets with PostGIS spatial queries and training an XGBoost model to rank geographic areas by health-risk exposure.

EDUCATION

Master of Science, Computer Science | Illinois Institute of Technology, Chicago, IL **Aug 2023 - May 2025**

Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Big Data Analytics, Advanced Database Organization, Cloud Computing, Data Mining, Natural Language Processing, Data Analysis, Computer Networks, Digital Healthcare Informatics and AI.

Bachelor of Technology, Computer Science and Engineering | Gautam Buddha University, India **Aug 2019 - Jun 2023**

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- IBM Applied AI Professional Certificate
- Codecademy: Data Scientist Professional
- Codecademy: Business Intelligence Data Analyst